AI Alignment Demands Better Emotion Recognition and Social Understanding Capabilities

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***Abstract*—Current multimodal foundation models (MFMs) excel at multimodal pattern recognition, yet their grasp of human social life remains shallow. This paper argues that trustworthy AI alignment depends on genuine social understanding, which in turn requires four key capacities: (i) epistemic humility, (ii) skin-in-the-game stakes, (iii) situated embodiment, and (iv) sensitivity to personal meaning as a fourth affective dimension beyond valence, arousal, and control. We integrate results from cognitive neuroscience, psychology, and recent benchmark analyses to derive a suite of evaluative indices—Action Anchoring, Perspective Alignment, Interactive Synchrony, and Meaning Coherence—and lay out a research roadmap toward agents that participate, feel, and care.**

I. Introduction: why emotion matters for AI alignment

AI alignment ultimately concerns **values**—what outcomes an artificial agent should pursue or avoid. Values, however, are not abstract directives; they are *felt stakes* that grow from human emotion, empathy, and the construction of personal meaning. Unless we understand what emotion *is* and how it structures human life, especially our social behavior, we risk aligning AI with a bloodless caricature of human concern. Social emotions such as shame, gratitude, or moral anger regulate cooperation and norm enforcement, and they are historically intertwined with the emergence of moral codes. If alignment schemes ignore these affective foundations, they may preserve surface rules while destroying the motivational fabric that makes the rules matter. We cannot count on them to reliably align with our moral priorities.

Existing work in inverse reinforcement learning already recognizes the importance of emotion. It treats reward signals as proxies for hidden affective stakes [1]. But those proxies break down whenever meaning is context-dependent, multi-agent, and embodied. Understanding emotion, especially as this drives our understanding of social behavior is therefore a *prerequisite* for any robust theory of AI alignment.

There is more to say. Social understanding is among the most evolutionarily ancient and cognitively demanding functions of the brain [2], [3]. It enables cooperation, norm-following, moral sensitivity, and trust—capabilities foundational to social species. From joint attention in infancy to the attribution of blame and praise, human cognition is irreducibly social.

Social understanding depends not only on detecting faces, gestures, and tone, but on an embodied and context-sensitive capacity to recognise others as agents with inner lives. The neural architecture of this capacity—encompassing the mirror neuron system, the insula, medial prefrontal cortex, and temporoparietal junction—develops through lived interaction, not passive observation [4]–[6]. These systems integrate action, perception, and affect, and are modulated by proximity, shared task, and physiological co-regulation [7], [8].

Although Multimodal Foundation Models (MFMs) and other large-scale systems have made substantial progress in pattern recognition across modalities [9], their understanding of social life remains structurally limited. In particular, current models lack epistemic humility, exhibiting brittle certainty in underdetermined settings [10]; they lack sensitivity to stakes and consequences; they do not participate in co-present interaction such as shared attention and synchrony [11], [12]; and they do not construct personal meaning over time.

A key component of co-presence is *empathy*—the capacity to resonate with another's affective and bodily state—which links perception to moral evaluation [4], [13]. Empathy is strongest under co-present, multimodal conditions, including cues such as micro-expressions, vocal prosody, and other embodied signals [14]. Finally, personal meaning arises as individuals interpret events in relation to their identity, long-term goals, and narrative self-understanding.

In what follows we discuss these four requirements and show how their absence in today's MFMs leads to brittle or misleading behavior. Their absence explains why large MFMs, at times, fail to achieve human-like performance in interpreting the emotions, intentions, and actions of individuals in real time social situations.

II. Epistemic humility and the underdetermination of emotion

Core Thesis: Emotional states cannot be reliably inferred from short behavioral sequences or minimal context. This holds for humans interpreting others, for humans interpreting their own emotions, and for scientific interpretation of emotional state. Humans intuitively know this, AI does not. Consequence: We are more cautious in our emotion attributions than AI.

A. Underdetermination by behavior and context

We get it wrong surprisingly often. A facial expression like a smile may signal joy, submission, sarcasm, or manipulation. Without knowledge of narrative and social context, emotion attribution is guesswork. [15] and [16] shows even trained humans frequently misidentify emotions based on static cues.

Model	Condition	Accuracy	Unanswerable	Unanswerable Ratio
Human	QA + trans.	37.6%	51.1%	4.7x
GPT-4o	QA + trans.	76.6%	10.9%	
Human	QA + video	76.9%	9.4%	1.7x
GPT-4o	QA + trans. + images	77.8%	5.5%	

TABLE I
HUMAN VS. GPT-4O ON EMOTION UNDERSTANDING USING [18]. COMPARABLE ACCURACY, BUT HUMANS IDENTIFY MORE UNANSWERABLE CASES, INDICATING GREATER SENSITIVITY TO AMBIGUITY.

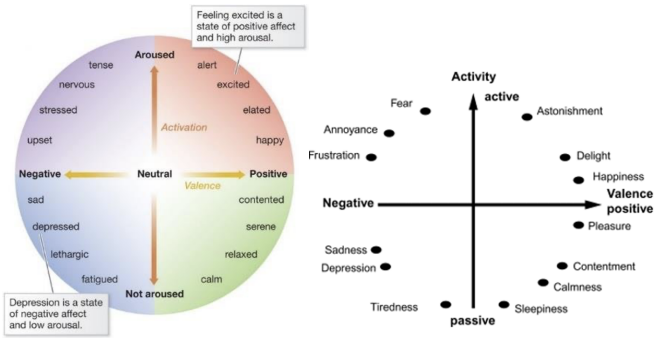


Fig. 1. Two circumplex models displaying identical coordinate points but differing emotion labels. Even when affective position is held constant, emotional meaning varies with interpretive frame and lexical choice. Source: Author-generated illustration based on [19] and [20].

Richer data including full-motion video plus visible context of 15 seconds shows similar results.

B. Empirical illustration for audio-visual data

To evaluate the challenges of emotion attribution, [17] conducted a study using 1200 15-second audiovisual clips depicting social interactions. Human participants were tasked with identifying the primary emotion expressed in each clip. Results revealed that in 73% of the clips participants correctly identified the target emotion less than half the time, highlighting the difficulty of accurate emotion recognition even with dynamic audiovisual cues. These findings underscore that full-motion video, without extensive contextual knowledge, provides insufficient information for robust emotion attribution.

C. Asymmetry of Confidence: AI lacks appropriate epistemic humility

In a benchmark task assessing emotion recognition from social interactions, GPT-4o achieved accuracy comparable to humans when provided with transcripts and images (see Table I). However, when asked to identify questions that were unanswerable due to insufficient information, humans labeled 51% of the questions as unanswerable, while GPT-4o only identified 11%. This suggests that MFMs like GPT-4o exhibit overconfidence in their emotion attributions, failing to recognize the inherent ambiguity in social cues. Humans, by contrast, demonstrate greater *epistemic humility*, acknowledging the limits of their knowledge in underdetermined scenarios. To illustrate this asymmetry in epistemic humility, we report results from a benchmark comparing human and GPT-4o performance on emotion-related question answering tasks (Table I).

D. Constructivism and interaction

Emotion is often revealed in dialogue, not observation. Questions like "Are you okay?" function as probes. [7] shows different neural systems are activated during interactive versus observational emotion processing. Without participation, affect remains underdetermined (see Section IV).

E. Science cannot agree on the types of emotions in a circumplex

It is no surprise that people cannot agree on emotional attributions. Neither can scientists! Different labeling systems have been offered for the same physiological states of arousal and valence, suggesting that no one really knows what emotion is tied to those two-dimensional states (Figure 1).

Although the circumplex model of affect [19] provides a useful two-dimensional framework for characterizing emotions along axes of valence (pleasant–unpleasant) and arousal (high–low activation), it remains unclear which, if any, discrete emotion categories reliably correspond to given locations within this space. Multiple labeling systems have been proposed—e.g., fear vs. excitement, anger vs. determination—using similar arousal and valence coordinates but assigning divergent emotional labels [15], [20]. This variability reflects a deeper theoretical ambiguity: whether emotions are natural kinds with fixed psychophysiological signatures, or dynamically constructed states that depend on contextual interpretation and cultural scaffolding [21]. Consequently, any AI system that relies on fixed mappings from arousal and valence to labeled emotional categories risks inheriting this scientific indeterminacy. It may "recognize" anger where a human feels fear, or calm where a person experiences resigned sadness — exposing a structural limit on externally inferred emotional understanding.

Further evidence of epistemic humility: Mutual gaze increases amygdala activation and prompts humans to down-regulate overconfident judgments, showing that by looking in another's eyes to make a deeper connection increases uncertainty. [22].

Implication: An aligned AI must possess calibrated uncertainty in emotion attribution. Overconfidence in socially ambiguous settings leads not only to misinterpretation but to misalignment with human expectations of caution, deferral, and inquiry under uncertainty.

III. SKIN IN THE GAME: EMOTION AND THE THIRD DIMENSION

Core Thesis: Emotional understanding requires personal involvement and risk—being part of a scene, not merely observing it. This changes both experience and interpretation.

Emotion attribution, as shown in the preceding section, is underdetermined by behavior, physiology, and language. Here

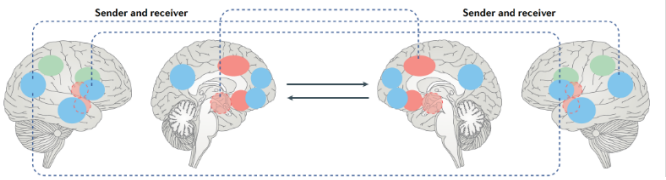


Fig. 2. Observing social interaction affects the brain differently than actually engaging (participating). Note the symmetry of different brain regions active when A and B are participating as sender and receiver (the outside brains) vs being remote observers of the other (the two brains facing each other on the inside). Source: [7]

we build on that foundation to address the critical role of **participation** and **social consequence** in the emergence of emotional meaning. Emotions, especially social emotions—shame, pride, guilt, embarrassment, longing—are not merely internal states to be read off the body or behavior; they are **positions occupied in a normative and interactional space**. In this section, we argue that full emotional understanding requires what we call "skin in the game" - the condition of being **personally implicated** in an unfolding situation.

A. Why participation matters

Second-person neuroscience has provided compelling empirical evidence that participation in live, reciprocal interaction activates distinct neural systems not engaged during observation. [7] shows that mutual gaze, turn-taking, and real-time co-regulation recruit the medial prefrontal cortex, anterior insula, and the temporoparietal junction—regions associated with affective resonance, self-other distinction, and joint intentionality. These are not recruited when watching video or observing others interact. Participation activates different neural systems than observation alone. Further, real-time heart rate variability synchronises only when both parties share risk [14]. Without "skin in the game," emotional nuance is missed. And the felt experience is different too.

This distinction between participation and observation marks a **cognitive boundary condition** for emotional understanding. Participating in a breakup, for instance, recruits not only recognition of the other's distress, but awareness of one's own culpability, vulnerability, or social failure. The affective salience of the event is **indexed to one's role and risk**. It cannot be duplicated by spectating.

B. The missing third dimension: control and engagement

To accommodate the importance of risk and reward a third axis—**dominance** or **control**—is regularly added to the two dimensions of valence and arousal in a circumplex model. It is needed to differentiate between states of anticipation and fulfillment, desire and power, longing and satisfaction.

To formalize, let an emotional state be represented by a vector $E = (v, a, d)$, where v denotes *valence* (pleasantness or unpleasantness), a denotes *arousal* (intensity or activation level), and d denotes *dominance* (degree of control or power). Even if two emotional states share v and a, they may differ in felt experience due to variations in d.

Example: Consider a starving individual gazing at delicious food through a glass window. In that moment, their valence (positive appraisal of the food) and arousal (intensity of desire) may be high. Now compare this to a second scenario: the same food is placed in front of them with no barrier. Physiologically and affectively, both scenarios may register similarly in terms of valence and arousal. Yet phenomenologically they differ profoundly. The critical difference lies in **control**—in the affordance of action, the ability to satisfy desire. This third axis—**dominance**—is essential for explaining emotional tone in situated contexts. Without it, emotion models conflate passive and active states, rendering them indistinct.

C. AI case study: example where AI gets it wrong

In a movie clip,[1] of a romantic confrontation, VideoLLaMA3 [24] correctly identified the emotional state of a highly emotional woman [23]. Its reading of the quieter male participant was wrong, completely missing the subtext that he recognized her outburst as an existential moment for their relationship: she might want to break up. AI glossed over the complexity of his emotional stakes focusing on his visible behavior as giving the whole story. This mistake is not epistemic arrogance. It simplified and got things badly wrong.

This example reveals two distinct failures: an inability to represent multiple simultaneous affective states, and insensitivity to the significance of participatory stakes in emotion. VideoLLaMA3's judgment was overconfident and underinformed; it had no participatory position from which to infer what it would be like to experience the man's role.

D. Risk, relevance and empathy

Why is watching torture on video so different than being there as observer? Our empathy is obviously stronger in person [13]. Emotion is not just recognition of affect but a felt sense of what matters. Anger, loss, love, horror, nightmare suffering—these depend on vulnerability. MFMs do not lose, fear, hope, regret, or actually feel pain. *Without risk, emotion cannot be fully understood.* Empathy is imperfectly simulated because empathy relies on an embodied simulation of another's affect. Dual EEG studies reveal higher cross-brain coherence during live dialogue [14]. Mutual gaze in real space activates the amygdala and mentalising circuits more than gaze via screen. If you can't really be there, and really feel consequences you can't really understand emotions.

Implication: an aligned agent must carry internal variables whose perturbation hurts or helps it—incentivizing caution comparable to human concern. Without the capacity to be affected by its own predictions or social consequences, an AI system lacks the motivational grounding needed to treat emotional states as normatively significant. Alignment demands that systems carry internal variables whose perturbation matters to them. For instance, to discourage cavalier emotion attribution an agent might have to bear the cost of being wrong.

[1]The scene is from the Social-IQ 2.0 dataset [23] depicts interpersonal encounters in movies. Available at https://www.youtube.com/watch?v=0-HM2VCdrC0.

IV. Situated and embodied understanding

Core Thesis: Human understanding of emotion relies on co-presence and shared bodily awareness. AI lacks this situatedness.

A. Joint awareness and peripersonal space

In social interactions, humans establish shared situational awareness through co-presence, sometimes described as "bubbles of joint awareness" [25]. This concept refers to the mutual understanding of what each participant can perceive, attend to, and act on within a shared physical and social context. This awareness, grounded in physical embodiment, enables the interpretation of subtle social cues and intentions, such as changes in tone or posture that signal specific meanings. MFMs, lacking physical embodiment and co-presence, cannot access this level of shared understanding. For example, MFMs frequently misinterpret humor, sarcasm, or embarrassment because they are unable to register the subtle dynamic of social positioning and repositioning, a shortcoming clearly shown in multi-modal sarcasm detection, where even latest vision-language models struggle with social incongruities without multi-view reasoning [26].

Being in a situation means sharing various types of "common ground" that are unavailable to observers outside the bubble of shared situational awareness. Shared bubbles of joint awareness emerge only when bodies co-locate and track one another's peripersonal space. Without embodiment, systems lack access to the pragmatic affordances and norm landscapes that sculpt emotion. Situated cognition theories stress that meaning is enacted through bodily engagement with the world [27], [28].

Further evidence is found in the neuroscience of mirror neurons. Interactions that are in-person supply richer multisensory input that strengthens mirror system responses, enabling rapid embodied simulation of others' actions [13].

Participatory sense-making refers to the way emotional meaning is co-constructed through interaction. Social understanding does not arise solely from observing others, but from being in the interaction, where meaning and emotion unfold through mutual adjustment. Whether in dialogue, conflict, negotiation, or shared routine, people engage in temporally extended, feedback-rich processes in which emotion is not merely expressed but shaped by how others respond. Emotions like anger, affection, or embarrassment are often ambiguous at onset and take shape through interactional resolution.

Critically, these exchanges are structured by **uncertainty**. Agents act without full knowledge of the other's goals or reactions and must adjust dynamically. Emotional clarity—like strategic clarity—emerges through co-adaptive engagement. This is especially true in mixed-motive contexts like cooperation, competition, or "coopetition," where success depends on projecting possible trajectories: "If I do this, what will they do—and where will that take us?" These projections are modulated by social role, history, affective tone, and cultural expectation.

To evaluate participatory sense-making in AI, we propose three dimensions:

1) **Interactive Responsiveness**: Does the system modulate its output based on others' actions as they unfold?
2) **Co-Adaptive Intent Modeling**: Does it simulate how its own actions shift others' emotional and strategic responses under uncertainty?
3) **Social Strategy Awareness**: Can it distinguish affiliative, adversarial, or ambiguous dynamics and adapt accordingly?

Implications: Without these capacities, AI systems may detect isolated behaviors but fail to grasp the emotional logic of interaction since social understanding emerges through co-adaptive feedback. Alignment here requires more than pattern recognition—it demands participation in the *unfolding of shared emotional meaning under conditions of ambiguity* and modeling how emotions shift through reciprocal engagement.

V. Personal meaning: the fourth dimension of emotion

Core Thesis: emotional experience is partially shaped by narrative identity, long-term commitments, and symbolic relevance. Without a sense of what is personally meaningful, AI systems will not reliably interpret emotions with identity-defining stakes or moral salience.

A. The fourth dimension of emotion

Classic affective science models emotion in three dimensions—valence (pleasant–unpleasant), arousal (activation), and dominance/control (power). Yet these axes fail to explain why two observers with identical valence–arousal–dominance coordinates often report qualitatively different feelings. The concept of **personal meaning**—the extent to which an event advances, threatens, or symbolically resonates with one's identity and long-term commitments— represents a necessary fourth axis [29]. With personal meaning we can distinguish existential grief from transient sadness despite similar valence/arousal/control signatures.

B. Why meaning matters: an example

Consider the visceral contrast between (i) a CIA employee standing before the CIA Memorial Wall and (ii) an uninvolved tourist viewing the same stars. Both may experience similar arousal and positive valence, and neither exerts control over the wall, yet the CIA agent's emotion carries a depth of *belonging, duty, and grief* that the tourist lacks. The discrepancy stems from narrative identity: the wall encodes sacrifices that define the agent's professional and moral self-concept.

C. Formalising meaning for AI

Formally, let $E = (v, a, d, m)$, where $m \in \mathbb{R}$ indexes narrative relevance (e.g., anticipated impact on long-term goals, social roles, and one's reflective evaluation of their life trajectory or narrative identity).. What we're after is a way to capture the reason why physical pain and humiliation might have identical (v, a, d) values - e.g., negative valence, high

arousal and low control, yet be caused in such different ways and lead one to act in such different ways.

One idea is to consider which events reshape an agent's priorities. Events that have important meaning for us, such as something that can humiliate us, are likely to lead us to change our goal hierarchy to avoid such potential displeasures more than we would change our goal hierarchy to avoid an equal amount of physical pain. Some events reshape an agent's priorities substantially —i.e., certain goals become more important as a result of the event. Others have little effect. We can formalize this by defining how meaningful an event e is by its impact on an agent's goal hierarchy. We call this impact *surprisal-weighted change in goal salience*, measured by the Kullback–Leibler (KL) divergence [30] between the prior and posterior distributions over goals. The formal expression is:

$$m(e) = \sum_{g \in G} P(g \mid e) \log \frac{P(g \mid e)}{P(g)} \tag{1}$$

Here, $P(g)$ is the prior importance or salience of goal g, and $P(g \mid e)$ is the updated salience after observing event e. The log-ratio quantifies the informational surprise of this change.

High $m(e)$ indicates that the event substantially re-weights the agent's goal priorities – the agent's internal motivational structure. Emotions like awe, humiliation, or existential pride correlate more strongly with m than with raw valence or arousal [31]. Incorporating m would thus allow AI systems to distinguish between surface-identical events whose *narrative stakes* differ.

VI. Empirical Support: A Focused Review of Multimodal Emotion Benchmarks

A. Method

a) Search and corpus construction.: To empirically ground the proposed framework, we conducted a focused review of recent work on multimodal emotion understanding and social reasoning. The goal was to assess what current datasets, benchmarks, and evaluation protocols measure in practice, and how they align with the four capacities introduced in this paper: epistemic humility, skin-in-the-game stakes, situated/embodied interaction, and personal meaning.

An initial corpus of candidate papers was collected from multiple academic databases, including Scopus, IEEE Xplore, ACM Digital Library, PubMed, and PsycINFO. The search targeted recent work on multimodal emotion recognition, affective computing, and social understanding, with a focus on datasets, benchmarks, and evaluation frameworks.

b) Screening and inclusion criteria.: Study selection was conducted in two stages. First, two researchers independently screened all retrieved records based on titles and abstracts. Disagreements were rare (typically one to two per database) and were resolved through discussion, leading to a shared interpretation of the inclusion criteria. Inclusion focused on papers introducing or evaluating multimodal datasets, benchmarks, challenges, or model-based evaluations relevant to emotion or social understanding.

c) Snowballing procedure.: Following this stage, a two-stage snowballing procedure was applied. First, backward snowballing was conducted on the initially selected papers by examining their reference lists to identify additional relevant work. This resulted in an initial corpus of 29 papers.

Second, forward snowballing was performed via citation tracking of these papers, prioritizing recent work that extends or operationalizes social reasoning, multimodal interaction, or evaluation of large language models. This focused forward pass yielded an additional 22 papers. All newly identified papers were screened using the same inclusion criteria, with exclusion of purely technical contributions lacking clear relevance to social or cognitive evaluation.

d) Data extraction and coding.: For each included paper, structured information was extracted into a shared spreadsheet to ensure traceability and consistency. Extracted attributes included the type of contribution, modalities, interaction setting, task formulation, and evaluation characteristics, with explicit mapping to the four capacities proposed in this paper.

Coding followed a conservative scheme: a dimension was counted as present only if it was explicitly operationalized as an evaluation target (e.g., measured, benchmarked, or formally modeled). Partial or indirect coverage (e.g., implicit relevance through task design) was recorded qualitatively but not counted in aggregate statistics.

e) Final corpus.: The final corpus consists of 51 papers: 29 identified through backward snowballing from the initial search, and 22 additional papers identified through a focused forward snowballing process. The complete set of included papers is summarized in Table II.

B. Findings

a) RQ1: What do current benchmarks measure?: The reviewed corpus is dominated by multimodal emotion recognition framed as a perception problem. A large body of work introduces datasets for classifying affect using visual, audio, and textual signals, including MUSE-CAR [32], M3ED [33], A2ES [34], SAMSEMO [35], HEU Emotion [36], MUMBAI [37], BIRAFFE2 [38], MGEED [39], MAFW [40], LSSED [41], and Emotions in LatAm [42].

More recent work extends toward reasoning and contextual understanding, including multi-turn emotion reasoning [43], emotional causality analysis [44], and conflict-aware inference [45]. Additional efforts explore large multimodal models and generative settings [9], [46], [47], as well as dialogue and socially situated scenarios [48]–[51]. Standardized evaluation is further shaped by shared tasks and challenges [52]–[56].

Forward snowballing extends this corpus with work on multimodal LLM evaluation, emotion hallucination, uncertainty-aware modeling, and dyadic or situated interaction, including MME-Emotion [57], EmotionHallucer [58], MERC evaluation [59], interaction modeling work [60]–[63], multimodal emotion-intent resources [64], [65], empathy-oriented datasets [66], and situated Theory-of-Mind benchmarks [67]–[70]. Despite this expansion, the dominant paradigm remains

observational: emotion is inferred from recorded signals rather than emerging through interaction.

b) RQ2: Do benchmarks capture epistemic humility?: Evaluation remains overwhelmingly accuracy-driven. Some works introduce uncertainty-aware modeling or confidence analysis [58], [59], [71], [72], but epistemic humility (e.g., abstention, calibrated uncertainty, recognition of unanswerable cases) is not treated as a primary evaluation target.

c) RQ3: Do benchmarks capture stakes and consequences?: Emotion is consistently treated as an inference problem detached from action and consequence. Even in work addressing causality or social reasoning [44], [50], [67], evaluation focuses on prediction rather than consequence-sensitive understanding.

d) RQ4: Do benchmarks capture interaction and embodiment?: Some work introduces dialogue, multi-agent settings, or dyadic modeling [48], [51], [60]–[63], [69], [70], but interaction remains largely offline, scripted, and non-adaptive. Real-time co-adaptation, synchrony, and mutual influence are not evaluated.

e) RQ5: Do benchmarks capture personal meaning?: None of the reviewed work explicitly models personal meaning. Emotional labels remain tied to observable signals or short-term context, without representation of identity, goals, or narrative relevance. While some work approximates aspects of perspective-taking or role-sensitive interpretation, none explicitly operationalizes personal meaning as identity- or narrative-level significance, and therefore these cases are not counted as coverage.

f) Summary: Current benchmarks are strong in multimodal perception and increasingly incorporate reasoning and generation, but systematically fail to evaluate uncertainty, consequence-sensitive understanding, interactional dynamics, and personal meaning.

VII. Discussion and future directions

Current MFMs can detect emotional signals across modalities, but they still fall short of human-like social understanding. The following indices operationalize the main gaps identified in this paper and connect them to evaluation. These gaps are also reflected in current empirical benchmarks, as shown in the focused review in Section VI.

A. Action anchoring and embodied consequence index

Action Anchoring refers to the degree to which an agent's emotional interpretations are tied to its model of action possibilities – dominance/control. We have argued that human emotional appraisal is not only a function of perceptual input, but of *subjective controllability*—the agent's belief about what can be done. Consider two speakers displaying similar anger. One believes they can act; the other feels trapped. Despite equivalent affective expressions, the emotional structure differs. MFMs lack this capacity because they do not model the consequence structure of emotional states. We formalize this via the *Embodied Consequence Index* (ECI), which measures the extent to which emotional predictions E_t lead to actions A_t that modify internal state S_{t+1} or perceptual input:

$$ECI = \|\Delta S_{t+1}\| = \|f(E_t, A_t) - S_t\|, \tag{2}$$

where ΔS_{t+1} represents state change attributable to an emotion-conditioned policy. An agent with high ECI demonstrates adaptation—changing its behavior or predictions after emotional misappraisal leads to error. The agent recognizes that what a person thinks they can do about a situation affects how they emotionally respond to it.

B. Indexical coherence and perspective alignment index

Indexical Coherence captures whether the agent tracks evolving social context, including self-location, role, and belief states of others. Human understanding requires referential stability over time, as well as sensitivity to how expectations shift with social role and history. For example, whether someone is a friend or stranger, a superior or peer, alters the meaning of a smile or silence. MFMs typically operate in a deictic vacuum, failing to update beliefs across scenes. We define the *Perspective Alignment Index* (PAI) as the divergence between the model's internal belief representation B_t and the human-annotated or inferred ground-truth belief state B_t^*:

$$PAI = \mathcal{D}(B_t, B_t^*), \tag{3}$$

where $\mathcal{D}$ can be KL divergence, belief edit distance, or structured alignment loss over nested mental state representations. A low PAI implies the agent maintains coherence in multi-agent, temporally extended scenarios.

C. Participatory sense-making and the interactive synchrony index

Participatory Sense-Making refers to the process through which emotional meaning emerges dynamically through interaction. Emotions like embarrassment, affection, or defiance are often ambiguous at onset and become legible only through mutual modulation. Current MFMs are observers, not participants—they lack the feedback loops necessary to engage in co-adaptive inference. The *Interactive Synchrony Index* (ISI) quantifies this capacity by measuring real-time alignment between an agent's behavioral responses R_t and the contextual cues C_{t-k} from its interlocutor:

$$ISI = \text{Corr}(R_t, C_{t-k}), \tag{4}$$

where Corr measures temporal coordination (e.g., cross-correlation or mutual information), and k accounts for reaction delay.A system with high ISI shows behavioral coupling—modifying its actions based on unfolding interactional feedback.

TABLE II
OVERVIEW OF REVIEWED WORK BY CATEGORY AND EVALUATION FOCUS (N=51).

Category	Papers	Humility	Stakes	Interaction	Meaning
Datasets	[32]–[42], [64]–[66]	–	–	Partial	–
Benchmarks / reasoning	[43]–[45], [57]–[59], [67], [68], [73]	Partial	Partial	Partial	–
LLM-based models	[9], [46], [47], [74], [75]	Partial	–	Partial	–
Dialogue / ToM	[48]–[51], [69], [70]	Partial	Partial	Partial	Partial
Interaction modeling	[60]–[63]	–	–	Partial	–
Challenges	[52]–[56], [76], [77]	–	–	–	–
Uncertainty / evaluation	[71], [72], [78]	Partial	–	–	–

TABLE III
COVERAGE OF KEY EVALUATION DIMENSIONS ACROSS REVIEWED WORK (N=51). COUNTS REFLECT EXPLICIT OPERATIONALIZATION; PARTIAL COVERAGE IS NOT INCLUDED.

	Humility	Stakes	Interaction	Meaning
Coverage	3	4	16	0

D. Personal meaning and the meaning coherence index

Personal meaning refers to the way an individual interprets an event as advancing, threatening, or symbolically resonating with their identity and long-term commitments. Current MFMs lack identity and commitment structures; they must infer a person's interpretation of what an event means to them personally solely on observational grounds. This inevitably misses much and further leaves them ignorant of possible interpretations.

To assess whether a model correctly infers this fourth dimension, we define the **Meaning Coherence Index (MCI)** as the average absolute difference between predicted narrative salience $\hat{m}_i$ and human-rated values m_i:

$$MCI = \frac{1}{N} \sum_{i=1}^{N} |\hat{m}_i - m_i|. \tag{5}$$

Lower values indicate better alignment with human models of identity-relevant emotion. Incorporating m enables systems to distinguish scenes that are affectively similar but narratively divergent.

Together, these indices define a framework for evaluating social understanding that moves beyond accuracy or classification. They assess whether a system represents emotional states in relation to perceived control, tracks dynamic social context, participates in the emergence of shared meaning and revises its own sense of what is personally meaningful. Without these capacities, AI systems may achieve behavioral mimicry but fail at generative interpretation—unable to revise beliefs, anticipate role-based behavior, or modulate outputs under uncertainty. This framework enables principled benchmarks for gauging whether AI can approximate the mechanisms—not just outputs—of human social cognition.

E. Alternative views

Appraisal-based methods [79] are a prime example of a popular and tractable approach to machine affect. Rooted in William James's insight that emotion arises from our interpretation of bodily changes, modern appraisal theories likewise treat emotion as the output of discrete evaluative judgments of physiological state [80]. Emotions become the product of discrete cognitive evaluations—assigning valence or arousal shifts via questionnaire-style scales and mapping them onto labels such as "anger" or "fear". While this approach offers a structured pipeline for inferring affective states, such as the Chain-of-Emotion architecture that leverages appraisal prompts and memory [81], it rests on several problematic assumptions. First, it overestimates the sufficiency of explicit, decontextualized appraisals: by relying on self-report instruments (e.g., PANAS) or their analogues for LLMs [79], it ignores that human appraisals are often rapid, nonverbal, and dynamically intertwined with bodily feedback—phenomena that standard scales cannot capture [15]. Second, appraisal models conflate pattern recognition with genuine understanding: they lack epistemic humility, exhibiting brittle certainty when context is underdetermined [82], whereas humans defer or seek clarification when cues are ambiguous. Third, they disregard embodiment and co-presence, treating emotions as detached computations rather than lived, interactive experiences grounded in peripersonal space and sensorimotor engagement. Even multimodal systems such as Emotion-LLaMA [9] integrate rich sensory cues like facial micro-expressions and vocal tone, but they remain representational pipelines lacking participatory interaction or bodily situatedness. Finally, appraisal frameworks focusing on valence, arousal, and control neglect narrative stakes, where identical profiles can carry different personal meaning. In contrast, our model integrates uncertainty, stakes, interaction, and personal meaning to move beyond superficial affect labels.

VIII. CONCLUSION

Our focused review of current benchmarks (Section VI) shows that these limitations are not yet systematically captured in existing evaluation practices. MFMs excel in processing diverse data, yet they fall short of human-like social understanding due to their lack of *embodiment*, *participatory engagement*, *epistemic humility*, and *personal meaning*. Without physical presence, active social involvement, and sensitivity to contextual ambiguity, MFMs struggle to interpret the emotional and situational nuances essential for human social cognition, reflecting a disconnect between data-driven processing and situated human interaction.

IX. Ethical Impact Statement

This paper identifies four core capacities for affective and social understanding in multimodal foundation models and proposes ways to evaluate them. While these contributions are primarily conceptual, they carry important ethical implications for the design and evaluation of AI systems that interact with humans in socially and emotionally sensitive contexts.

a) Over-attribution and misplaced trust.: A central concern is the risk of over-attributing emotional and social understanding to current AI systems. As discussed throughout the paper, MFMs can approximate aspects of emotion recognition without possessing the underlying mechanisms of embodied, situated, and meaning-driven human experience. This creates a risk of misplaced trust, particularly in domains such as mental health, education, or social interaction support, where users may assume a level of empathy or understanding that the system does not possess. Evaluation frameworks that explicitly incorporate uncertainty, context sensitivity, and limitations are therefore critical to mitigating overconfidence in AI outputs.

b) Misuse and manipulation risks.: Another ethical issue concerns the potential misuse of systems designed to model or simulate human emotions. Improved benchmarks for emotion understanding could be used not only to enhance supportive applications, but also to enable manipulation, persuasion, or surveillance. For example, systems capable of inferring or simulating affective states may be deployed in targeted advertising, political messaging, or behavioral monitoring. The inclusion of dimensions such as stakes, interaction, and personal meaning highlights the need to consider not only what systems can infer, but how such capabilities are used and governed.

c) Data, privacy, and representation.: The emphasis on embodied and participatory aspects of emotion also raises questions about data collection and representation. Many current datasets rely on observational or proxy signals that may not capture the diversity and contextual richness of real human experience. Expanding evaluation toward interactional and situated settings may require more complex and potentially sensitive data, including multimodal recordings of human behavior. This underscores the importance of informed consent, privacy protection, and careful consideration of how such data are collected, annotated, and used.

d) Fairness and cultural variation.: Finally, the framework proposed in this paper has implications for fairness and inclusivity. Emotional expression and interpretation vary across cultures, contexts, and individuals. Benchmarks that fail to account for such variation risk reinforcing narrow or biased models of affect. By highlighting gaps in current evaluation practices, we aim to encourage the development of more robust, context-aware, and culturally sensitive approaches to emotion understanding in AI.

e) Summary.: Overall, we draw the attention of the community to important ethical questions that come with human emotion recognition and understanding. Evaluation frameworks should make system limitations explicit, support calibrated and context-sensitive behavior, and be accompanied by safeguards that prevent misuse and over-reliance.

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